

Cognitive Computing

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What is CC [Revise] :

- “Cognitive computing represents self-learning systems that utilize machine learning models to mimic the way brain works.” Eventually, this technology will facilitate the creation of automated IT models which are capable of solving problems without human assistance”

3 essential components of a cognitive computing system:

- **A way of interpreting**
- **A body of content/information that supports the decision**
- **A way of processing the signal against the content/info *corpus***





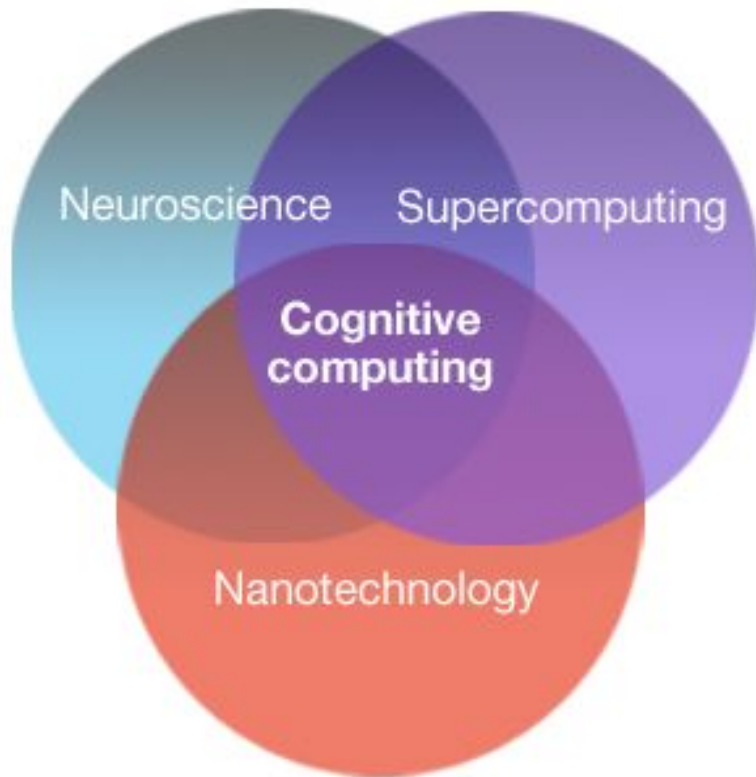
Run Run ..Its fire!
I have to save my life..
Oh God!
Save my life!!

Fire? I don't feel any
fire here!!



- ✓ When there is a fire the tries to come out from the room.. but the super computer fails at that point..
- ✓ So what is great?
- ✓ Obviously!! Yeah... you heard it right!!
- ✓ So it is necessary to make computers think like Human(Animal).

Introduction



- **What Is Cognition?**
- Mental action of acquiring knowledge through thoughts, experience and senses.

What is Cognitive Computing?

- *“Technology for the development of computer system from s/w and h/w elements which have ability to think and react like human minds.”*
- Combination of **Neuroscience** , **Super computing** and **Nanotechnology**.

Why it is too hard for computer to understand humans?

Structured Data

Unstructured Data

Where was Einstein born?

Physicist	Birth Place
A. Einstein	Ulm
N. Bohr	Copenhagen
M. Curie	Warsaw

Source: Spreadsheet, Database, etc.

"One day, from among his city views of Ulm, Otto chose a water color to send to Albert Einstein as a remembrance of Einstein's birthplace"

Source: <http://www.schaeffenacker-ulm.de/en/otto.html>

Welch ran this?

Person	Organization
L. Gerstner	IBM
J. Welch	GE
W. Gates	Microsoft

Source: Spreadsheet, Database, etc.

"If leadership is an art then surely Jack Welch has proved himself a master painter during his tenure at GE"

Source: *Jack Welch and the GE Way*, Robert Slater

Need Of Cognitive Computing:

- Today's Information Challenge:



Data volume is expanding at an incredible rate

...Data will grow 800% in the next five years

...Unstructured data growing faster than structured



Data is getting more social

...20M articles on Wikipedia

...30B pieces of Facebook content are shared monthly

...There are 156M public blogs



An estimated 2 billion people are now on the Web

... and a trillion connected objects – cars, appliances, cameras, roadways, pipelines

Search vs. Expert Q&A

Decision Maker

Has Question

Distills to 2-3 Keywords

Reads Documents, Finds Answers

Finds & Analyzes Evidence

Search Engine

Finds Documents containing Keywords

Delivers Documents based on Popularity

Decision Maker

Asks NL Question

Considers Answer & Evidence

Expert

Understands Question

Produces Possible Answers & Evidence

Analyzes Evidence, Computes Confidence

Delivers Response, Evidence & Confidence

What if an enterprise has all the answer it need to succeed?



Can we design a computing system that rivals a human's ability to retrieve, analyze and interpret vast amounts of information?

Purpose:

- *“to create a new generation of technology that can find answers from data more effectively than current search engines”*

IBM Watson

- For this purpose, the very first cognitive computer named “Watson” was developed by IBM co.
- Capable of answering questions posed in natural language
- Watson was named after IBM's Thomas J. Watson.

IBM Watson(cont.)

- Search engines deliver thousands of results that match keywords
- University's have worked on a consistent question answering software for years
- Programmed by **25 IBM scientists**
- Not connected to the internet



Jeopardy! Challenge!

- February 2011: **First computer to compete against humans in Jeopardy!**
- Defeated shows greatest two champions Ken Jennings and Brad Rutter



Technology

- Question-answering Technology
- Deep understanding of **natural language**.
- Computer running Software called Deep QA
 - Holds approximately one million books worth of information

Examples Of Cognitive Technology

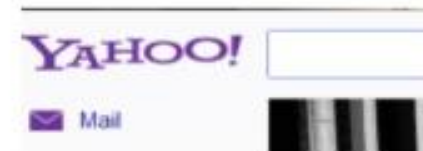
Examples of Cognitive Computing



- Web Search



- Image Search



- Event Search



- Social Computing

Customers Who Bought This Item Also Bought

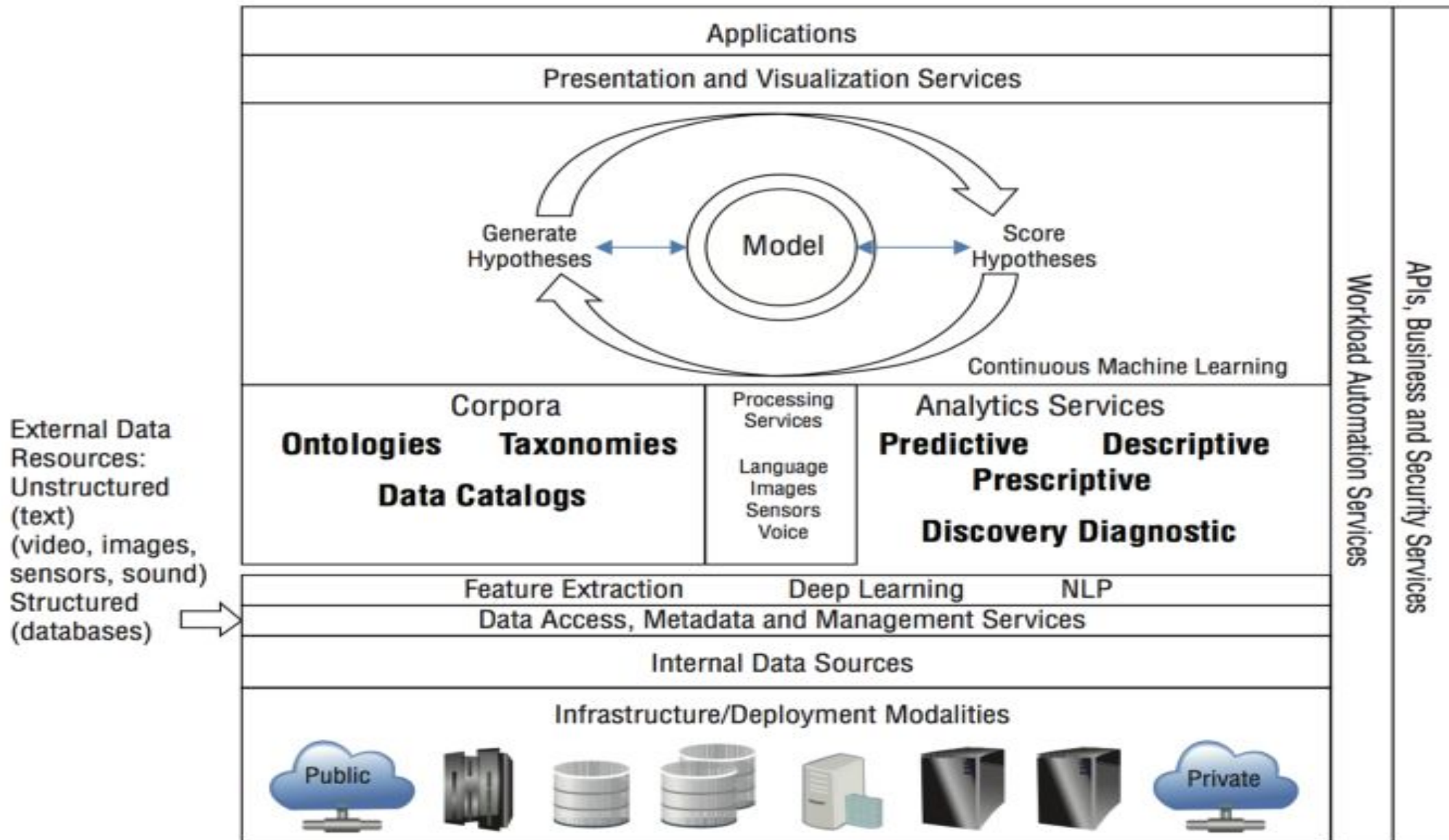


- Natural Language Processing



IBM Corporation

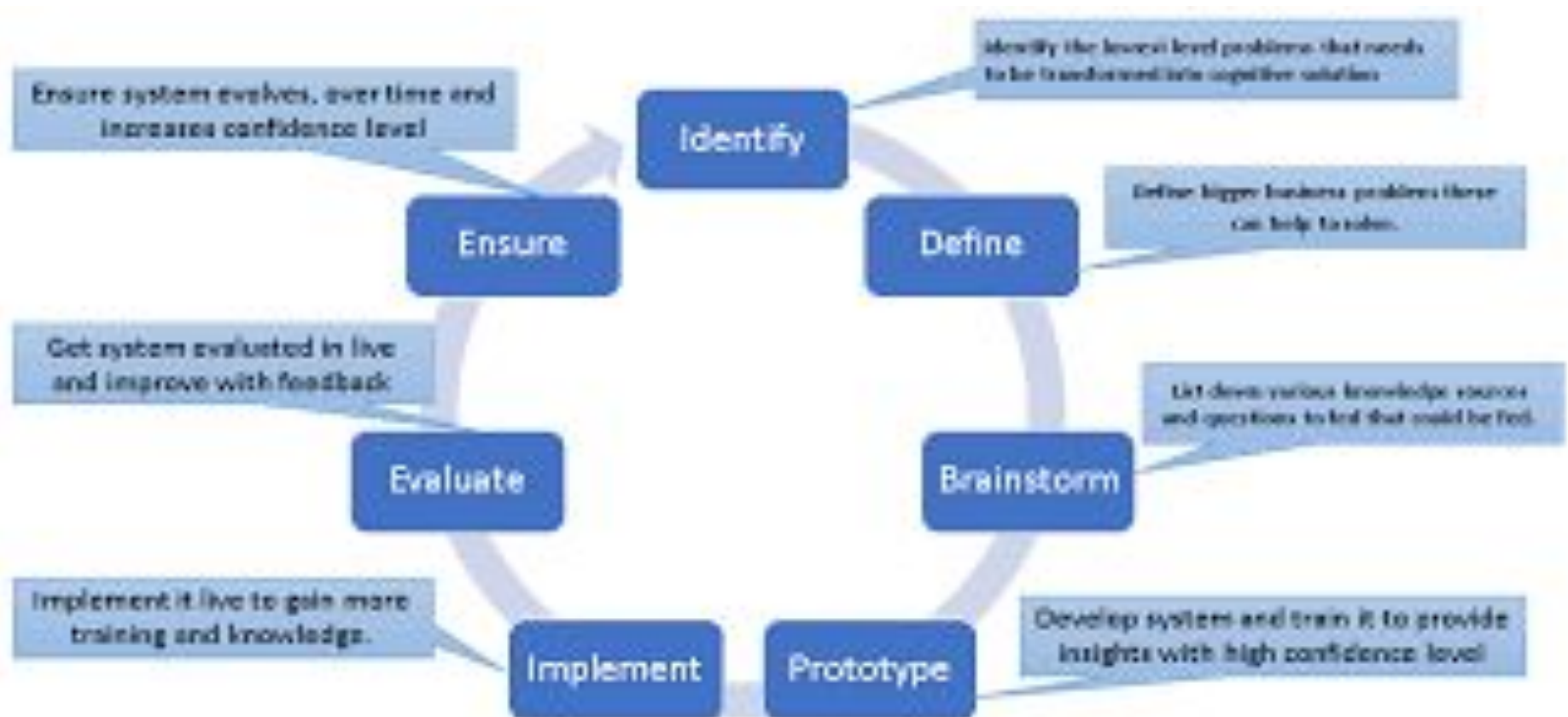
Elements of Cognitive Computing



.....Elements of CC

- **Infrastructure**
- **Data Access , Metadata & Management Services**
- **Corpus, Taxonomies & Data Catalogs**
- **Data Analysis Services**
- **Continuous ML**
- **Hypothesis Generation and Evaluation**
- **Tools and Learning Process**
- **Presentation and Visualization**
- **Cognitive Application**

Design principles for CS



Identify: lowest level problem

Define: Define bigger business problems

Brainstorm : knowledge source and questions to be fed

prototype: develop a system

Implement

Evaluate :

Ensure : System evolves

NLP in support of a Cognitive system

- Natural language processing (NLP) is a core ability of cognitive computing systems and is often defined as helping computers process and understand human language

Phonetics and phonology	The study of language sounds
Word Sense Disambiguation(WSD)	Defined as the ability to determine which meaning of word is activated by the use of a word in a particular context
Morphology	The study of meaningful components of words
Syntax	The study of structural relationships among words
Lexical semantics	The study of word meaning
Compositional semantics	The study of the meaning of sentences
Pragmatics	The study of the use of language to accomplish goals
Discourse conventions	The study of conventions of dialogue

Discipline	Typical Problems	Tools
Linguists	How do words form phrases and sentences? What constrains the possible meanings for a sentence?	Intuitions about well-formedness and meaning; mathematical models of structure and meaning
Psycholinguists	How do people identify the structure of sentences? How are word and text meanings identified?	Experimental techniques based on measuring human performance; statistical analysis of observations
Philosophers	What is meaning, and how do words and sentences acquire it? How do words identify objects in the world?	Natural language argumentation using intuition about counter-examples; mathematical models (for example, logic and model theory)
Computational Linguists	How is the structure of sentences identified? How can knowledge and reasoning be modeled? How can language be used to accomplish specific tasks?	Algorithms, data structures; formal models of representation and reasoning; AI techniques (search and representation methods)

Phonology/ Phonetics

Phonology is the study of speech sounds used in a particular language

(read)

Morphology

is the study of the structure of words/formation of words

..unfairness

..unhappiness

Syntax

Structure of the language

How words are grouped to form sentences

Semantics

meaning of the words in a sentence and how these words are combined to form meaningful sentences

“ That Company is facing the huge financial crises and in May it may vary”

Pragmatics

situational use of the language ... Different from semantics....

“Do not judge a book by its cover...”

Discourse

Group of sentences and it studies or finds actual meaning of the context by connecting component sentences

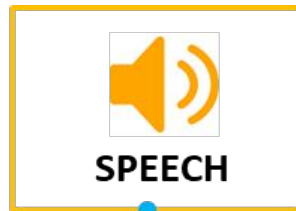
“ I took a book from college library . Then went to Project Lab and I left the book there

Word Sense Disambiguation (WSD)

.....ability to determine which meaning of word is activated by the use of word in particular context

The chicken is ready to eat.

Is the chicken ready to eat his food or the chicken is ready for someone else to it? You never know.



Phonological analysis

OCR/Tokenization

Pragmatic analysis 6

Meaning of sentence in various situation
– real world knowledge

Disclosure Integration 5

Focusing on properties of text as a whole and
making connections between sentences

Semantic analysis 4
Checking Meaningfulness



Morphological analysis 1

Primitive unit of meaning

Lexical analysis 2

Interpreting the meaning of
individual words

Syntactic analysis 3
Uncovering the grammatical structure of sentences

Phonological Analysis: This level is applied only if the text origin is a speech. It deals with the interpretation of speech sounds within and across words. Speech sound might give a big hint about the meaning of a word or a sentence.

[analyzing the sound]

Morphological Analysis: Deals with understanding distinct words according to their morphemes (the smallest units of meanings) . Taking, for example, the word: “**unhappiness**”. It can be broken down into three morphemes (prefix, stem, and suffix), with each conveying some form of meaning: the prefix **un-** refers to “**not being**”, while the suffix **-ness** refers to “**a state of being**”. The stem happy is considered as a free morpheme since it is a “word” in its own right. Bound morphemes (prefixes and suffixes) require a free morpheme to which it can be attached to, and can therefore not appear as a “word” on their own.

Lexical Analysis: It involves identifying and analysing the structure of words. Lexicon of a language means the collection of words and phrases in a language. Lexical analysis is dividing the whole chunk of txt into paragraphs, sentences, and words. In order to deal with lexical analysis, we often need to perform **Lexicon Normalization**.

The most common lexicon normalization practices are Stemming:

Stemming: Stemming is a rudimentary rule-based process of stripping the suffixes (“ing”, “ly”, “es”, “s” etc) from a word.

Lemmatization: Lemmatization, on the other hand, is an organized & step by step procedure of obtaining the root form of the word, it makes use of vocabulary (dictionary importance of words) and morphological analysis (word structure and grammar relations).

Syntactic Analysis: Deals with analyzing the words of a sentence so as to uncover the grammatical structure of the sentence. E.g.. "Colourless green idea." This would be rejected by the Symantec analysis as colourless here; green doesn't make any sense.

Semantic Analysis: Determines the possible meanings of a sentence by focusing on the interactions among word-level meanings in the sentence. Some people may think it's the level which determines the meaning, but actually all the level do. The semantic analyser disregards sentence such as “hot ice-cream”.

Discourse Integration: Focuses on the properties of the text as a whole that convey meaning by making connections between component sentences. It means a sense of the context. The meaning of any single sentence which depends upon that sentences. It also considers the meaning of the following sentence. For example, the word "that" in the sentence "He wanted that" depends upon the prior discourse context.

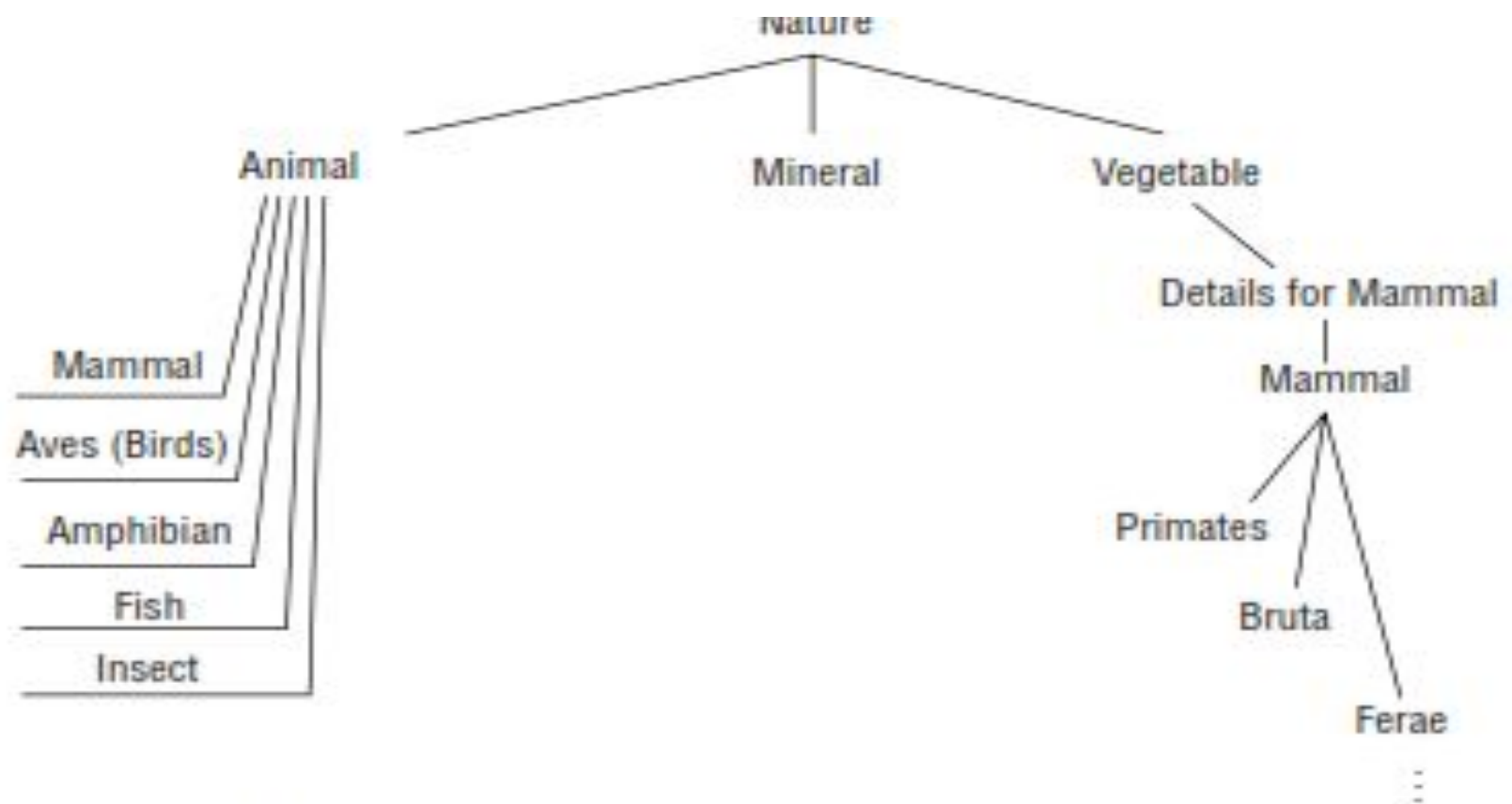
Pragmatic Analysis: Explains how extra meaning is read into texts without actually being encoded in them. This requires much world knowledge, including the understanding of intentions, plans, and goals. Consider the following two sentences:

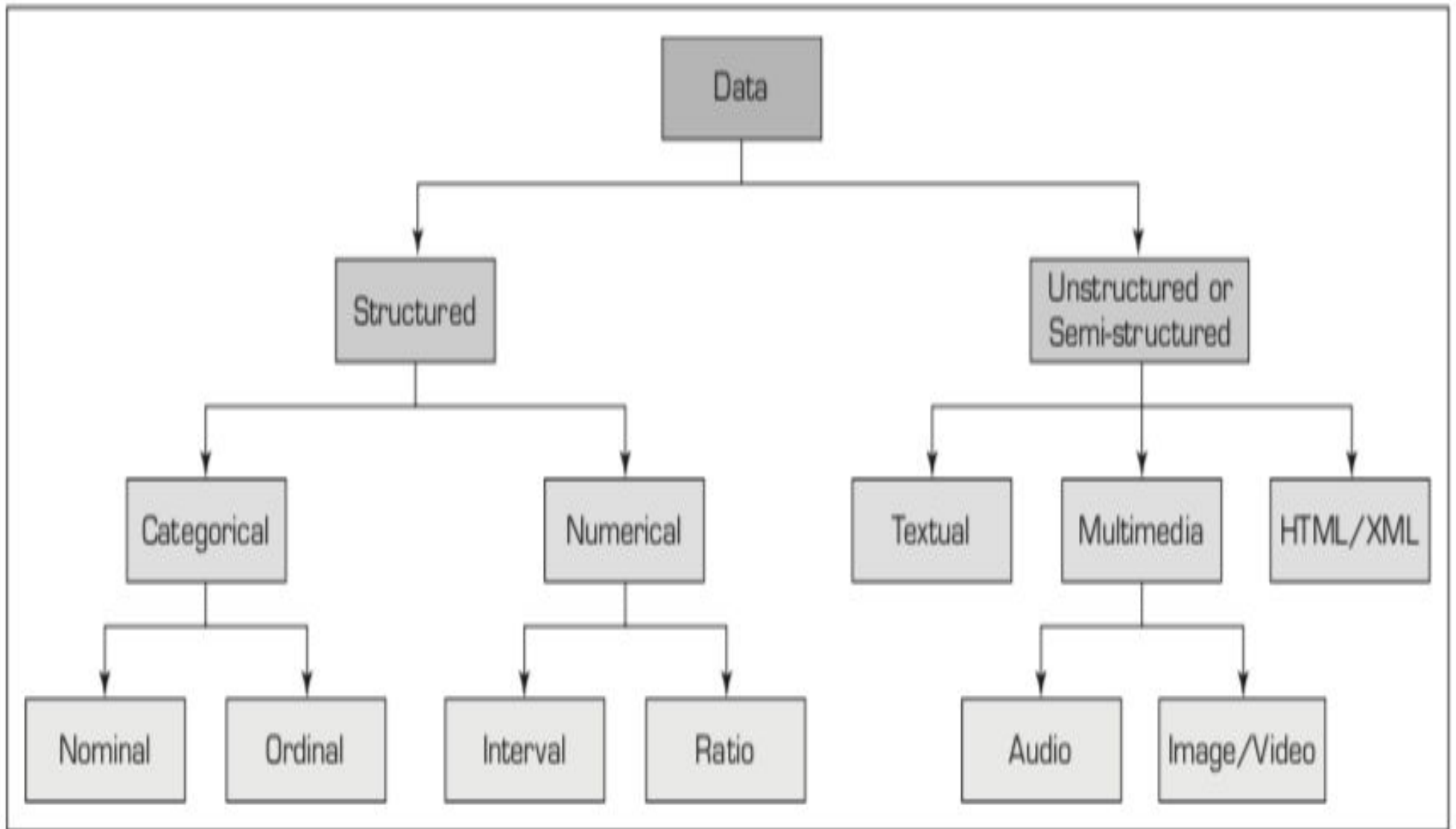
- The city police refused the demonstrators a permit because they feared violence.
- The city police refused the demonstrators a permit because they advocated revolution.

Representing Knowledge in Taxonomies and Ontologies

Taxonomies

A taxonomy is a representation of the formal structure of classes or types of objects within a domain. Taxonomies are generally hierarchical and provide names for each class in the domain.





Ontologies[the branch of metaphysics dealing with the nature of being.]

An ontology provides more detail than a taxonomy, although the boundary between them in practice is somewhat fuzzy. An ontology should comprehensively capture the common understanding vocabulary, definitions, and rules—of a community as it applies to a specific domain

Anatomy of Ontologies

- Class
- Relationship
- Attribute

TAXONOMY VERSUS ONTOLOGY

Taxonomy refers to a branch of science in which the organisms are defined and named based on their shared characteristics

Anatomical, physiological & evolutionary study of organisms

Uses the hierarchy of the organisms to describe and classify organisms

Involved in naming, defining, and classifying organisms

A comparatively simple phenomenon

Produces a tree with branches like relationships

Ontology refers to a branch of science that studies the nature of being, its existence or reality as well as the hierarchy of the being

A type of philosophical study about organisms

Uses more complex variations such as the nature of the being, its reality

Involved in classification of organisms using a complex and sophisticated model where everything is interconnected

More complex than taxonomy

Produces a web-like relationship where everything is interconnected

Methods of Knowledge representation

- Simple Tree
- Semantic Web

Applying advance Analytics to Cognitive Computing

ANALYTICS TYPE	DESCRIPTION	EXAMPLES OF QUESTIONS ANSWERED
Descriptive Analytics	Understand what happens when using analytic techniques on historical and current data.	Which product styles are selling better this quarter as compared to last quarter? Which regions are exhibiting the highest/lowest growth? What factors are impacting growth in different regions?
Predictive Analytics	Understand what might happen when using statistical predictive modeling capabilities, including data mining and machine learning. Predictive models use historical and current/real-time data to predict future outcomes. Models look for trends, clusters of behavior, and events. Models identify outliers.	What are the predictions for next quarter's sales by product and region? How does this impact raw material purchases, inventory management, and human resource management?
Prescriptive Analytics	Use to create a framework for making a decision about what to do or not do in the future. The "predictive" element should be addressed in prescriptive analytics to help identify the relative consequences of your actions. Use an iterative process so that your model can learn from the relationship between actions and outcomes.	What is the best mix of products for each region? How will customers in each region react to advertising promotions and offers? What type of offer should be made to each customer to build loyalty and increase sales?

Prescriptive Analytics	Use to create a framework for making a decision about what to do or not do in the future. The "predictive" element should be addressed in prescriptive analytics to help identify the relative consequences of your actions. Use an iterative process so that your model can learn from the relationship between actions and outcomes.	What is the best mix of products for each region? How will customers in each region react to advertising promotions and offers? What type of offer should be made to each customer to build loyalty and increase sales?
Machine Learning and Cognitive Computing	Collaboration between humans and machines to solve complex problems. Assimilate and analyze multiple sources of information to predict outcomes. Need depends on the problems you are trying to solve. Improve effectiveness of problem solving and reduce errors in predicting outcomes.	How secure is the city environment? Are there any alerts from the vast amount of information streaming from monitoring devices (video, audio, and sensing devices for smoke or poisonous gases)? Which combination of drugs will provide the best outcome for this cancer patient based on the specific characteristics of the tumor and genetic sequencing?

Analytics Maturity level

Q: How much did we sell as a company ?

Q: How many customers churned in last last quarter?

Q: what was our overall productivity ?

Analytics Maturity level

Q : why did Company sales decrease in the previous quarter ?

Q: Why are we seeing an increase in customer churn ?

Q : why is a specific basket of products vastly outperforming their year sales figures ?

Analytics Maturity level

Q: Determine credit risk and identifying potential fraud

Q : predict and avoid customer churn by identifying signs of customer dissatisfaction

Analytics Maturity level

Q : Automatic adjustment of products pricing based on anticipated customer demand and external factors

Some NLP Applications

finding appropriate documents on certain topics from a database of texts (for example, finding relevant books in a library)

extracting information from messages or articles on certain topics (for example, building a database of all stock transactions described in the news on a given day)

translating documents from one language to another (for example,
producing automobile repair manuals in many different languages)

summarizing texts for certain purposes (for example, producing a 3-page summary of a 1000-page government report)

Some more NLP Applications

question-answering systems, where natural language is used to query a database (for example, a query system to a personnel database)

automated customer service over the telephone (for example, to perform banking transactions or order items from a catalogue)
tutoring systems, where the machine interacts with a student (for example, an automated mathematics tutoring system)

spoken language control of a machine (for example, voice control of a VCR or computer)

Dave: Open the pod bay doors, HAL.
HAL: I am sorry, Dave. I am afraid I can't do that.
Dave: What's the problem.
HAL: I think you know what the problem is just as well as I do.
Dave: I don't know what you're talking about.
HAL: I know that you and Frank were planning to disconnect me, and I'm afraid that's something I cannot allow to happen.

General speech and language understanding and generation capabilities

Politeness: emotional intelligence

Self-awareness: a model of self, including goals and plans

Belief ascription: modeling others; reasoning about their goals and plans

Production-Level Applications

A computer program in Canada accepts daily weather data and automatically generates weather reports in English and French

Over 1,000,000 translation requests daily are processed by the Babel Fish system available through Altavista

A visitor to Cambridge, MA can ask a computer about places to eat using only spoken language. The system returns relevant information from a database of facts about the restaurant scene.

Prototype-Level Applications

Computers grade student essays in a manner indistinguishable from human graders

An automated reading tutor intervenes, through speech, when the reader makes a mistake or asks for help

A computer watches a video clip of a soccer game and produces a report about what it has seen

A computer predicts upcoming words and expands abbreviations to help people with disabilities to communicate

Stages in a Comprehensive NLP System

Tokenization

Morphological Analysis

Syntactic Analysis

Semantic Analysis (lexical and compositional)

Pragmatics and Discourse Analysis

Knowledge-Based Reasoning

Text generation

The bane of NLP: ambiguity

Ambiguity resolution at all levels
and in all system components is
one of the major tasks for NLP

Translation

The coach lost a set

One strongly preferred meaning although
in a standard English-Russian dictionary

coach has 15 senses

lose has 11 senses

set has 91 sense

$15 \times 11 \times 91 = 15015$ possible translations

Noise in the communication channel

hte

Easily resolvable

But sometimes, it is less clear:

Thanks for all you help!

This sentence is ambiguous: It has a reading as is; but it can also be misspelled...

How does one process this?

Conclusion

- We believe that cognitive systems will help us make sense of our amazingly complex world and provide a new collaborative dimension to human reasoning and decision making.
- These technologies will help create new products, services, companies and entire industries.

THANK YOU 🤗